EE3621: Digital Signal Processing

Lecture 15: Power Spectral Density & Wiener Filter
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Random Signals Review: WSS definition, autocorrelation properties.
- **05:00 – 12:00 (7 mins):** Power Spectral Density (PSD): Wiener-Khinchin theorem and detailed proofs.
- **12:00 – 15:00 (3 mins):** Cross-correlation and Cross-PSD definitions.
- **15:00 – 22:00 (7 mins):** LTI systems with random inputs: Output autocorrelation and PSD proofs, white noise concepts.
- **22:00 – 30:00 (8 mins):** Wiener Filter formulation: Orthogonality principle, derivation of Wiener-Hopf equations.
- **30:00 – 35:00 (5 mins):** Causal solution, Minimum Mean Square Error (MMSE) and Application examples.
- **35:00 – 40:00 (5 mins):** Checkpoints and detailed worked examples.

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2 Random Signals Review

In many real-world Digital Signal Processing (DSP) applications, signals are not deterministic. Instead, they are modeled as random processes.

2.1 Wide-Sense Stationary (WSS) Processes

A random process $x[n]$ is Wide-Sense Stationary (WSS) if it satisfies two conditions:

1. The mean is constant: $E\{x[n]\} = \mu_x$ for all n.
2. The autocorrelation depends only on the time difference (lag) m, not on absolute time n.

2.2 Autocorrelation Sequence

For a WSS process, the autocorrelation sequence is defined as:

$$R_{xx}[m] = E\{x[n]x^*[n-m]\} \tag{1}$$

where $E\{\cdot\}$ denotes the expected value and $*$ denotes complex conjugation.

Key Properties:

1. **Average Power:** $R_{xx}[0] = E\{x[n]x^*[n]\} = E\{|x[n]|^2\}$.
2. **Conjugate Symmetry:** $R_{xx}[-m] = R_{xx}^*[m]$.
3. **Maximum Value:** $|R_{xx}[m]| \leq R_{xx}[0]$.

3 Power Spectral Density (PSD)

The Power Spectral Density (PSD), $S_{xx}(e^{j\omega})$, describes how the power of a random signal is distributed across different frequencies.

3.1 Wiener-Khinchin Theorem

Theorem: The PSD of a WSS process is the DTFT of its autocorrelation sequence:

$$S_{xx}(e^{j\omega}) = \sum_{m=-\infty}^{\infty} R_{xx}[m]e^{-j\omega m} \tag{2}$$

Proof: Let $x_N[n]$ be a truncated version of $x[n]$. The DTFT is $X_N(e^{j\omega}) = \sum_{n=-N}^{N} x[n]e^{-j\omega n}$. The expected value of the periodogram is:

$$E\{P_N(e^{j\omega})\} = \frac{1}{2N+1}E\left\{\sum_{n=-N}^{N} x[n]e^{-j\omega n} \sum_{k=-N}^{N} x^*[k]e^{j\omega k}\right\} \tag{3}$$

$$= \frac{1}{2N+1}\sum_{n=-N}^{N}\sum_{k=-N}^{N} E\{x[n]x^*[k]\}e^{-j\omega(n-k)} \tag{4}$$

Let $m = n - k$. Grouping terms with the same lag gives:

$$E\{P_N(e^{j\omega})\} = \sum_{m=-2N}^{2N}\left(1 - \frac{|m|}{2N+1}\right)R_{xx}[m]e^{-j\omega m} \tag{5}$$

Taking the limit as $N \to \infty$ proves the theorem:

$$S_{xx}(e^{j\omega}) = \lim_{N\to\infty} E\{P_N(e^{j\omega})\} = \sum_{m=-\infty}^{\infty} R_{xx}[m]e^{-j\omega m} \tag{6}$$

3.2 Properties of PSD

- **Real and Non-Negative:** $S_{xx}(e^{j\omega})$ is real and $S_{xx}(e^{j\omega}) \geq 0$.
- **Average Power:** $R_{xx}[0] = \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{xx}(e^{j\omega})d\omega$.

4 Cross-Correlation & Cross-PSD

For two WSS processes $x[n]$ and $y[n]$:

$$R_{xy}[m] = E\{x[n+m]y^*[n]\} \tag{7}$$

$$S_{xy}(e^{j\omega}) = \text{DTFT}\{R_{xy}[m]\} = \sum_{m=-\infty}^{\infty} R_{xy}[m]e^{-j\omega m} \tag{8}$$

5 LTI System with Random Input

For an LTI system $y[n] = h[n] * x[n]$.

5.1 Output Autocorrelation

$$
\begin{aligned}
R_{yy}[m] &= E\{y[n]y^*[n-m]\} && (9)\\
&= E\left\{\left(\sum_k h[k]x[n-k]\right)\left(\sum_l h^*[l]x^*[n-m-l]\right)\right\} && (10)\\
&= \sum_k \sum_l h[k]h^*[l]R_{xx}[m+l-k] && (11)\\
R_{yy}[m] &= h[m] * h^*[-m] * R_{xx}[m] && (12)
\end{aligned}
$$

5.2 Output PSD

Taking the DTFT:

$$S_{yy}(e^{j\omega}) = |H(e^{j\omega})|^2 S_{xx}(e^{j\omega}) \tag{13}$$

5.3 White Noise

White noise $v[n]$ has flat PSD: $R_{vv}[m] = \sigma_v^2\delta[m]$ and $S_{vv}(e^{j\omega}) = \sigma_v^2$.

6 Wiener Filter

We observe $x[n] = s[n]+v[n]$ and want to estimate desired signal $d[n]$ using $\hat{d}[n] = \sum_k w[k]x[n-k]$.

6.1 Orthogonality Principle & Wiener-Hopf Equations

To minimize $E\{|e[n]|^2\}$, the error must be orthogonal to inputs: $E\{e[n]x^*[n-l]\} = 0$.

$$E\left\{\left(d[n] - \sum_k w[k]x[n-k]\right)x^*[n-l]\right\} = 0 \tag{14}$$

$$R_{dx}[l] - \sum_k w[k]R_{xx}[l-k] = 0 \tag{15}$$

$$\sum_k w[k]R_{xx}[l-k] = R_{dx}[l] \tag{16}$$

In matrix form for FIR filters: $\mathbf{R}\mathbf{w} = \mathbf{r}$.

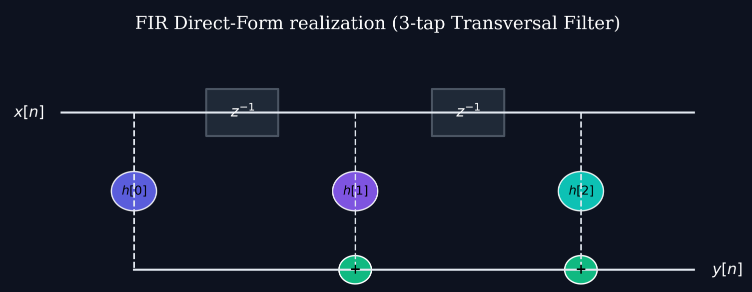

Figure 1: FIR direct-form implementation for Wiener filter.

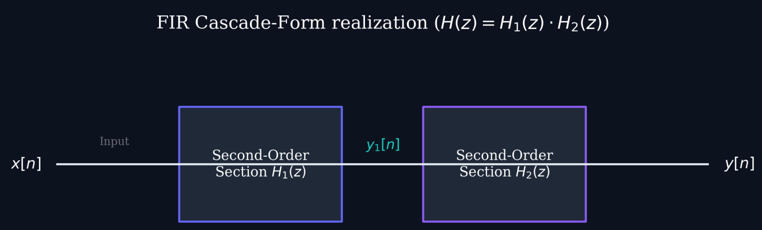

Figure 2: Alternative FIR cascade-form structure.

6.2 Non-Causal Solution and MMSE

In frequency domain:

$$W_{opt}(e^{j\omega}) = \frac{S_{dx}(e^{j\omega})}{S_{xx}(e^{j\omega})} \tag{17}$$

The Minimum Mean Square Error (MMSE) is:

$$\xi_{min} = R_{dd}[0] - \mathbf{w}^T \mathbf{r} \tag{18}$$

7 Checkpoints

Q1: PSD for $R_{xx}[m] = a^{|m|}$.
Solution: $S_{xx}(e^{j\omega}) = \sum a^{|m|} e^{-j\omega m} = \frac{1-a^2}{1-2a\cos\omega + a^2}$.
Q2: 2-tap Wiener Predictor for $R_{xx} = [1, 0.8, 0.5]$.
Solution: Solve $\mathbf{Rw} = \mathbf{r}$: $w_0 + 0.8w_1 = 0.8$, $0.8w_0 + w_1 = 0.5$. Yields $\mathbf{w} = [1.1111, -0.3889]^T$.
Q3: MMSE for Q2.
Solution: $\xi_{min} = 1 - (1.1111(0.8) - 0.3889(0.5)) = 0.3056$.